MIL-M-23813A(SHIPS)
21 December 1964
SUPERSEDING
MIL-M-23813(SHIPS)
12 December 1963
(See 6.4)

MILITARY SPECIFICATION

MICROPHONES, INTERCOMMUNICATION AND ANNOUNCING SYSTEMS, NAVAL SHIPBOARD

1. SCOPE

1.1 Scope.- This specification covers microphones designed for use in intercommunication and announcing systems aboard Naval ships. The microphones are noise canceling types suitable for use in noisy areas.

1.2 Classification.- Microphones shall be of the following types, as specified (see 6.2):

Type

M124/SIC - Hand held, equipped with 18 inch length of MCOS-5 cable and six pin connector (replacement for type IC/MPM-3).
M125/SIC - Hand held, equipped with 120 inch coil cord and six pin connector (replacement for types IC/MPM-6 and M-114/U).
M132/SIC - Hand held, equipped with 120 inch coil cord and six pin connector (replacement for type IC/MPM-4).
M131/WIC - Hand held, equipped with 120 inch coil cord and ten pin connector (replacement for type M-106/U).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL
QQ-N-281 - Nickel-Copper-Alloy (Monel and R-Monel) Bars, Plates, Rods, Sheets, Strips, Wire, Forgings, and Structural and Special Shaped Sections.
QQ-B-750 - Bronze, Phosphor, Bar, Plate, Rod, Sheet, Strip, Flat Wire and Structural and Special Shaped Sections.

MILITARY
MIL-I-631 - Insulation, Electrical, Synthetic-Resin Composition, Nonrigid.
MIL-S-901 - Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements For.
MIL-C-915 - Cable, Cord and Wire Electrical (Shipboard Use).
MIL-I-983 - Interior Communication Equipment, Naval Shipboard, Basic Design Requirements For.
MIL-C-10544 - Connector, Plug and Receptacle, Electrical, Audio, Waterproof, Ten Contact, Polarized.
MIL-E-17555 - Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of.
MIL-E-21981 - Electronics Type Designations, Identification Plates and Markings, Requirements For.

STANDARDS

MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

FSC 5830

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DRAWINGS

BUREAU OF SHIPS
9000-S6502 74117 - Microphone, Plug and Receptacle.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies should be addressed to Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, New York.)

AMERICAN STANDARDS ASSOCIATION (ASA)
Z24.8-1949 - Laboratory Standard Pressure Microphones, Specification For.
S1.6-1960 - Acoustical Measurements, Preferred Frequencies For.

(Application for copies should be addressed to the American Standards Association, 10 East 40th Street, New York 16, New York.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Preproduction sample.- Prior to beginning production two samples shall be tested as specified in 4.2 (see 6.3 and 6.3.1).

3.2 Description.- Microphones shall be hand held types. Each microphone shall consist of either a dynamic or magnetic element and a press-to-talk switch housed in a plastic case with a microphone cable and connector attached as specified herein. Unless otherwise specified (see 6.2), a microphone holder assembly as specified in 3.11 shall be furnished with each microphone.

3.3 General Features.- In addition to the requirements specified herein, the microphones shall be in accordance with the following paragraphs of MIL-I-983. Whenever a requirement of MIL-I-983 conflicts with a requirement of this specification, the requirement of this specification shall govern:

Fungus - inert materials
Toxic materials
Wood
Metals
Protection against corrosion
Rounded corners and edges
Drilled and tapped holes
Extended storage
Gunblast (see 3.12.13)
Shock, vibration and inclination (see 3.12.15 and 3.12.16)
Soldering
Workmanship and general examination

3.4 Form factor.- The general configuration of the microphone case shall be such that it fits well in the hand. The microphone case shall fit and lock into the microphone holder assembly specified in 3.11.

3.5 Plastic material.- The plastic material used in making the microphone case shall be of such quality that it will withstand all of the tests specified herein without evidence of physical damage. Plastic material shall be black.

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3.6 Replaceable parts.- The microphone design shall allow easy replacement of the press-to-talk switch and the microphone cable.

3.7 Cables and connector plugs.- Microphones shall be equipped with cables and connectors as specified in 3.7.1 through 3.7.3. The cable shall be secured by a clamp in the microphone case to prevent strain on the microphone electrical terminals.

3.7.1 Microphone type M124/SIC.- The type M124/SIC microphone shall be equipped with an 18-inch length of type MCOS-5 cable in accordance with MIL-C-915 and a microphone connector plug of the type shown on Drawing 9000-S6502-74117.

3.7.2 Microphone types M125/SIC and M132/SIC.- Type M125/SIC and M132/SIC microphones shall be equipped with a 120-inch length (extended) of retractable cordage conforming to the general requirements of MIL-C-915 for MCOS-5 cable, except that it shall be coiled. (Koiled Kords, Inc., type 4230, or equivalent is considered acceptable.) Type M125/SIC and M132/SIC microphones shall also be provided with a microphone connector plug of the type shown on Drawing 9000-S6502-74117.

3.7.3 Microphone type M131/WIC.- The type M131/WIC microphone shall be equipped with a microphone cable of the type specified in 3.7.2 and a type U-77/U microphone connector plug in accordance with MIL-C-10544.

3.8 Press-to-talk switch.- The press-to-talk switch shall be a positive action double pole single throw switch and shall be capable of closing the control circuit when actuated. It shall be mounted in such a manner that the pushbutton is at the top of the microphone.

3.9 Electrical circuit.- The electrical circuit of microphone type M132/SIC shall conform to figure 1. The electrical circuit of microphone types M124/SIC and M125/SIC shall be in accordance with figure 2. The electrical circuit of microphone type M131/WIC shall be as shown on figure 3.

3.10 Wiring diagram.- A printed card, tag, or label showing a schematic wiring diagram of the microphone, cord, and plug with all parts named for rapid and easy identification, shall be supplied with each microphone.

3.11 Microphone holder assembly, type MT-3400/SIC.- Each microphone holder assembly shall be assembled in accordance with figure 4 from the following individual parts:

(a) Microphone clamp.- One microphone clamp as shown on figure 5 made of annealed cold rolled Monel with satin finish in accordance with QQ-N-281.
(b) Spring.- One spring as shown on figure 6 made of phosphor bronze material in accordance with QQ-B-750 and nickel plated or of copper-nickel-alloy material in accordance with QQ-N-281.
(c) Catch.- One catch as shown on figure 7 made of annealed cold rolled Monel in accordance with QQ-N-281.
(d) Rivets.- Four stainless steel or nickel plated brass rivets 0.121 inch diameter and 5/32 inch length.

3.12 Performance requirements.-

3.12.1 Frequency response.- The frequency response shall fall within the limits shown by the solid lines on figure 8 when measured in accordance with 4.6.1. The zero decibel (db) reference level shall be the microphone output voltage measured at a frequency of 1000 cycles per second (c.p.s.).

3.12.2 Microphone output voltage.- The output voltage measured in accordance with 4.6.2 shall be a minimum of 1 millivolt root mean square (r.m.s.).

3.12.3 Microphone linearity.- The difference between the frequency response curves measured in accordance with 4.6.3 shall be within 1.5 db. of 20 db. throughout the frequency range from 200 to 6400 c.p.s.

3.12.4 Noise canceling.-

3.12.4.1 Preproduction samples.- The microphone shall have an Articulation Index of 0.5 or greater when determined as specified in 4.6.4.1.

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3.12.4.2 Production samples.- The on-axis to off-axis voltage output ratios for production samples shall not differ by more than 1.5 db. from the corresponding ratios measured for the preproduction sample used as a standard when subjected to the tests specified in 4.6.4.2.

3.12.5 Induced voltage.- When the microphone is subjected to the induced voltage tests specified in 4.6.5.1 and 4.6.5.2, the measured induced voltage shall not exceed 20 microvolts in either test.

3.12.6 Microphone endurance.- The microphone shall be capable of withstanding the endurance test specified in 4.6.6 without evidence of physical or electrical damage.

3.12.7 Switch endurance.- The microphone press-to-talk switch shall be capable of completing 100,000 operations when tested in accordance with 4.6.7 without evidence of mechanical or electrical failure.

3.12.8 Cable assembly strength.- There shall be no evidence of physical damage to any of the electrical connections following the cable assembly strength test specified in 4.6.8.

3.12.9 Dielectric strength.- The electrical circuits shall be capable of withstanding the dielectric strength test specified in 4.6.9.

3.12.10 Insulation resistance.- Insulation resistance shall be greater than 10 megohms when measured in accordance with 4.6.10.

3.12.11 Extended storage.- The microphone shall be capable of withstanding the extended storage test specified in 4.6.11 without evidence of any physical damage.

3.12.12 Salt spray.- The microphone shall be capable of withstanding the salt spray test specified in 4.6.12 without evidence of physical damage or corrosion.

3.12.13 Gunblast.- The microphone shall be capable of withstanding the gunblast test specified in 4.6.13 without evidence of physical damage.

3.12.14 Random drop.- The microphone shall be capable of withstanding the random drop test specified in 4.6.14 without evidence of physical damage.

3.12.15 Vibration.- The microphone shall be capable of withstanding the vibration test specified in 4.6.15 without evidence of physical damage.

3.12.16 Shock.- Following the shock test specified in 4.6.16 the microphone shall be capable of normal operation and shall show no evidence of physical damage.

3.12.17 Frequency response after environmental test.- The resultant frequency response curve shall fall within the limits defined by the dotted lines shown on figure 8. The output voltage at 1000 c.p.s. measured in the test performed after the environmental tests specified in 4.6.11 through 4.6.16 shall be used as the zero db. reference level in determining conformance to 4.6.17. The output voltage at the 1000 c.p.s. shall not have changed more than 3 db. from the original value as a result of the series of environmental tests.

3.13 Equipment identification.-

3.13.1 Identification of microphones.- Each microphone furnished under this specification shall be marked to show the microphone nomenclature and the number of the contract under which the microphone is procured. Microphones shall be marked by means of a cable band-marker of polyvinylchloride material in accordance with MIL-I-631.

3.13.2 Identification of microphone holder.- The microphone holder assembly shall have the designation "Microphone Holder MT-3400/SIC" stamped on the side of the piece shown on figure 4.

3.14 Nomenclature.- Government assigned nomenclature is required for microphones furnished under this specification. The nomenclature assigned to each type microphone will be essentially that used in this specification except a letter may be added after the number portion of the nomenclature to indicate a specific design. The supplier shall be required to request Government nomenclature in accordance with MIL-E-21981.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction inspection.- One sample shall be subjected to all of the examination and tests listed in table I which shall be performed, in general, in the order listed, and shall meet the requirements of this specification prior to manufacture of the production units. One sample, which has been subjected to only the general examination, frequency response, microphone output voltage, microphone linearity, noise canceling, and induced voltage tests listed in table I and shown to meet the requirements of this specification for those tests shall be marked and returned to the supplier for use as a standard reference in performance of the noise canceling test required by 4.6.4.2 for production samples.

Table I - Preproduction inspection.

Inspection	Requirement paragraph	Test paragraph
General examination	3.3	4.5
Frequency response	3.12.1	4.6.1
Microphone output voltage	3.12.2	4.6.2
Microphone linearity	3.12.3	4.6.3
Noise canceling	3.12.4.1	4.6.4.1
Induced voltage	3.12.5	4.6.5
Microphone endurance	3.12.6	4.6.6
Switch endurance	3.12.7	4.6.7
Cable assembly strength	3.12.8	4.6.8
Dielectric strength	3.12.9	4.6.9
Insulation resistance	3.12.10	4.6.10
Extended storage	3.12.11	4.6.11
Salt spray	3.12.12	4.6.12
Gunblast	3.3 and 3.12.13	4.6.13
Random drop	3.12.14	4.6.14
Vibration	3.3 and 3.12.15	4.6.15
Shock	3.3 and 3.12.16	4.6.16
Frequency response	3.12.17	4.6.17

4.3 Sampling for quality conformance inspection.-

4.3.1 Inspection lot.- All microphones presented for delivery at one time shall be considered a lot. The lot may include the entire contract quantity or it may be the production of any convenient time period.

4.3.2 Sampling for group A examination and tests.- A sample of microphones shall be selected from each inspection lot in accordance with table II and subjected to the group A examination and tests specified in 4.4 (table IV). The results of each test shall be compared with specification requirements. Failure to conform to this specification for any group A examination or test shall be counted as a defect, and the microphones shall not be offered for delivery. If the number of such nonconforming microphones in any sample exceeds the acceptance number for that sample, this shall be cause for rejection of the lot represented by that sample.

4.3.3 Sampling for group B tests.- A sample of microphones shall be selected from each inspection lot in accordance with table III and subjected to the group B tests specified in 4.4 (table IV) with lot acceptance based on the sampling requirements of MIL-STD-105. The results of each test shall be compared with specification requirements. Failure to conform to the specification requirements for any group B test shall be counted as a defect, and the microphones shall not be offered for delivery. If the number of such nonconforming microphones in any sample exceed the acceptance number for that sample, this shall be cause for rejection of the lot represented by that sample.

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Table II - Sampling for group A examination and tests
AQL (approx.) = 1.5 percent defective.

Number of microphones in inspection lot	Number of microphones in sample	Number of microphones nonconforming on group A examination and tests	
		Acceptance number	Rejection number
7 and under	all	-	-
8 to 15	7	0	1
16 to 40	10	0	1
41 to 110	15	0	1
111 to 300	25	1	2
301 to 500	35	1	2
501 and over	50	2	3

Table III - Sampling for group B tests.

Number of microphones in inspection lot	Number of microphones in sample	Number of microphones nonconforming on group B tests	
		Acceptance number	Rejection number
3 and under	all	-	-
4 to 15	3	0	1
16 to 40	5	0	1
41 to 110	7	0	1
111 to 300	10	0	1
301 to 500	15	1	2
501 and over	25	2	3

4.3.4 Sampling for group C tests.- Group C tests will be required by the bureau or agency concerned when the basic design of the microphone or the material of a vital part has been changed. Three microphones shall be selected and subjected to the group C tests specified in 4.4 (table IV). The results of each test shall be compared with specification requirements. In the event of failure to conform to this specification for any group C test, the supplier shall correct the cause of failure on future production units and repair the deficiency in all microphones produced on the contract or order.

4.4 Quality conformance inspection.- The sample microphones selected in accordance with 4.3 shall be subjected to the examination and tests listed in table IV. Tests shall be performed, in general, in the order listed.

4.5 General examination.- The general examination shall be performed in accordance with MIL-I-983.

4.6 Test procedures.-

4.6.1 Frequency response.-

4.6.1.1 Acoustic environment.- The acoustic environment in which the frequency response and microphone linearity tests are made shall be free from reflected waves that would affect the measured results by more than one db. at any test frequency. The maximum permissible acoustic noise level of the environment shall be such that the output which it produces is at least 20 db. below that produced during a test made in accordance with 4.6.1.4.

4.6.1.2 Sound source.- The sound source used for the frequency response and microphone linearity tests shall be an artificial voice constructed as shown on figure 9. The artificial voice consists of a loudspeaker driver unit, acoustically coupled to a hemispherical baffle. The throat is acoustically damped to provide a

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smooth response in the 180-5600 c.p.s. pass band. The artificial voice shall be driven by a suitable power amplifier and the amplifier-voice combination shall be capable of producing an r.m.s. sound pressure level of 80 microbars at a point on-axis and 1/2 inch from a plane across the front opening of the artificial voice for any test frequency in the range from 180 to 5600 c.p.s. with a maximum distortion of 5 percent. The sine-wave signal to be supplied to the power amplifier shall be obtained from a continuously variable frequency sine-wave generator. Means shall be provided to keep the sound pressure level constant within plus or minus 2 db. throughout the frequency range from 180 to 5600 c.p.s. This may be achieved by using an automatic level control operating from a probe microphone in the throat of the artificial voice, an electrical equalizing network, or an adjustable gain control coupled to the sine-wave generator.

Table IV - Quality conformance inspection.

Inspection	Requirement paragraph	Test paragraph
Group A		
General examination	3.3	4.5
Frequency response	3.12.1	4.6.1
Microphone output voltage	3.12.2	4.6.2
Group B		
Microphone linearity	3.12.3	4.6.3
Noise canceling	3.12.4.2	4.6.4.2
Induced voltage	3.12.5	4.6.5
Dielectric strength	3.12.9	4.6.9
Insulation resistance	3.12.10	4.6.10
Group C		
Microphone endurance	3.12.6	4.6.6
Switch endurance	3.12.7	4.6.7
Cable assembly strength	3.12.8	4.6.8
Extended storage	3.12.11	4.6.11
Salt spray	3.12.12	4.6.12
Gunblast	3.3 and 3.12.13	4.6.13
Random drop	3.12.14	4.6.14
Vibration	3.3 and 3.12.15	4.6.15
Shock	3.3 and 3.12.16	4.6.16
Frequency response	3.12.17	4.6.17

4.6.1.3 Calibration.- The sound pressure field produced by the artificial voice shall be calibrated by means of a standard microphone. The standard microphone shall be a pressure operated microphone that meets the requirements of ASA Publication Z24.8-1949, for type L. The use of types M or H microphones is permissible providing that a baffle is added to simulate the physical size of type L. The standard microphone shall be mounted in front of the artificial voice in such a position that its axis is colinear with that of the voice. The microphone face shall be 1/2 inch from the front surface of the artificial voice. The pressure calibration of the standard microphone shall be used in calibrating the sound source.

4.6.1.4 Test procedure.- The test microphone shall be so positioned that the axis normal to the plane of its opening is colinear with the axis of the artificial voice. The distance from the microphone face to the front surface of the artificial voice shall be 1/2 inch. The microphone audio circuit shall be terminated with a 150 ohm noninductive resistor. The voltage on the voice coil of the artificial voice shall be adjusted to the value required to produce a sound pressure of 20 microbars at the face of the test microphone with the signal frequency set at 1000 c.p.s. The test microphone output voltage shall be measured with a laboratory type vacuum tube voltmeter. This measured voltage shall be the zero db. reference level for the microphone frequency response. The audio signal frequency shall be swept through the 180 to 5600 c.p.s. range and the test microphone output in decibels referenced to the 1000 cycle level shall be recorded, using audio frequency semilogarithmic graph paper. The resulting response curve shall be compared with figure 8 to determine compliance with 3.12.1.

4.6.2 Microphone output voltage.- The microphone output voltage shall be the voltage measured across the 150 ohm resistor with a 1000 c.p.s. 20 microbar sound pressure applied to the microphone as specified in 4.6.1.4. Results shall comply with 3.12.2.

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4.6.3 Microphone linearity.- The frequency response measurement described in 4.6.1.4 shall be repeated for each of two conditions. For the first test the reference sound pressure level at 1000 c.p.s. shall be 5 microbars. For the second test the reference sound pressure level at 1000 c.p.s. shall be 50 microbars. The frequency response at both levels shall be recorded on a single sheet of semilogarithmic paper. The results shall comply with 3.12.3.

4.6.4 Noise canceling test.-

4.6.4.1 For preproduction samples.-

4.6.4.1.1 Artificial head voice.- The close sound source used in these measurements shall be the one designated "AAF Type I Korogel Head," described in Response Characteristics of Interphone Equipment, Report OSRD No. 3015 of 1 January 1944, Cruft Laboratory, Harvard University, Cambridge, Massachusetts, or its equivalent. (This consists of an "average" sized Plaster of Paris head coated with Korogel, a plastic which gives it approximately the consistency of human flesh; a loudspeaker driver unit is mounted behind the head and coupled to the lips by a 3/4 inch diameter tube. This head shall be mounted on a clothed dummy resembling the upper part of the human torso in its acoustic properties.)

4.6.4.1.2 Diffuse sound field.- A diffuse random noise field shall be provided in a region sufficiently large to contain the artificial head (and dummy), and such that the sound pressure 2 feet in front of the artificial head is not changed in any octave band by more than 0.5 db. upon complete removal of the head (and dummy) from the region. The sound pressure and first order pressure gradient shall be constant to within 1 db. with respect to direction and position over a 5 cubic foot region containing the artificial head, measured in the absence of the head, in each octave band.

4.6.4.1.3 Signal source.- The signal, both for the artificial head voice and diffuse random noise field shall be thermal noise. Sufficient equalization shall be provided so that the spectrum level will not vary more than 2 db. in any octave band used for the test.

4.6.4.1.4 Filters.- A General Radio Company, type 1558-AP Octave-Band Noise Analyzer or a Bruel and Kaer Model 2112 One-Third and Full Octave Band Analyzer shall be used in performing the noise canceling test. The octave bands shall be centered at 250, 500, 1000, 2000 and 4000 cycles which are the preferred frequencies listed in ASA publication S 1.6-1960.

4.6.4.1.5 Calibration.- Both the artificial head voice and the diffuse random noise field shall be calibrated in each band with a probe microphone. For calibration of the artificial head voice the probe microphone shall be positioned with its axis at a right angle to the axis of the voice. The tip of the probe microphone shall be placed 1/4 inch from a plane across the front of the lips which is perpendicular to the voice axis. The microphone to be tested shall not be mounted in the normal test position during the calibration of the noise field and the voice. For calibration of the diffuse random noise field the probe microphone shall remain in the position used for calibration of the voice and the artificial head and dummy shall be removed.

4.6.4.1.6 Measurement of random noise discrimination.- The true r.m.s. value of the sound pressure and microphone output voltage shall be measured in each case. After calibrating the sound sources as described in 4.6.4.1.5, the noise canceling microphone shall be mounted in the normal position of use on the artificial head. The point of the microphone closest to the center of the mouth shall be 1/4 inch from the plane of the "lips" or as close thereto as is consistent with the term "normal position of use." The microphone audio circuit shall be terminated with a 150 ohm non-inductive resistor. The increase in microphone sensitivity to the artificial voice signal, over its sensitivity to the diffuse random noise (each sensitivity defined as the ratio in decibels, reference 1 volt/microbar, of the output voltage to the sound pressure measured in 4.6.4.1.5) shall be measured in each octave band.

4.6.4.1.7 Calculation of articulation index.- The value of the Articulation Index (A.I.) shall be calculated by inserting the measured values of C; and the appropriate values of W; and Q; from table V into the formula:

$$A.I. = \sum_{i=1}^{5} \frac{W_i}{30} \left\langle C_i + Q_i \right\rangle$$

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where C_i is the random noise discrimination (expressed in decibels) in the ith octave band (i = 1, 2, 3, 4, and 5); W_i is the articulation index weighting factor for each band; and Q_i are the constants for each band listed in table V. The angular brackets indicate that if the quantity within them is less than zero or greater than 30, it is to be taken as zero or 30, respectively.

Table V - Calculation of articulation index.

Band No. i	Center frequency (cycles per second)	Limits (cycles per second)	W_i	Q_i
1	250	180-355	.072	7.5
2	500	355-710	.136	10.1
3	1000	710-1400	.224	8.8
4	2000	1400-2800	.318	4.7
5	4000	2800-5600	.227	- 1.4

4.6.4.2 For production samples.- Production samples shall be tested as specified in 4.6.4.2.2 and the test results, expressed as specified in 4.6.4.2.3, compared with those obtained for the preproduction sample subjected to the same test to determine compliance with 3.12.4.2.

4.6.4.2.1 Preproduction sample as a standard.- One of the preproduction samples which has been tested at a Government laboratory and meets the requirements of 3.12.1, 3.12.2, 3.12.3 and 3.12.4.1 shall be used as a standard microphone to which production samples shall be compared. The preproduction sample used for a standard shall not have been subjected to any of the environmental tests required by 3.12.11 through 3.12.16.

4.6.4.2.2 Test procedure.- The sound source for this test shall be the artificial voice specified in 4.6.1.2. The artificial voice output shall be adjusted to produce a sound pressure level of 50 microbars at a point on-axis and 1/2 inch in front of the voice for test frequencies of 400 and 1000 c.p.s. The microphone output voltage shall be measured across a 150 ohm resistive load for each of the following test conditions.

4.6.4.2.2.1 Close talking on-axis.- The microphone shall be mounted in the normal test position as specified in 4.6.1.4.

4.6.4.2.2.2 Distant talking off-axis.- The microphone shall be positioned in front of and facing the artificial voice with its principal axis parallel to that of the voice. The position of the microphone shall be such that the center of its face is at a point 3 inches from the center of the front opening of the voice on a line drawn at a 45 degree angle to the principal axis of the artificial voice.

4.6.4.2.3 Ratio of on-axis to off-axis voltage output.- The ratio of output voltage measured with the microphone on-axis as specified in 4.6.4.2.2.1 to the output voltage measured with the microphone off-axis as specified in 4.6.4.2.2.2 shall be expressed in decibels for each of the two test frequencies.

4.6.5 Induced voltage.-

4.6.5.1 Electromagnetically induced voltage.- The microphone to be tested shall be complete with microphone cable and connector plug. A 150 ohm noninductive resistor shall be connected to the microphone audio circuit at the connector end of the cable. The press-to-talk switch shall be closed and a 0.5 ampere r.m.s. 60 c.p.s. sinusoidal current made to flow through the control circuit leads. The induced voltage shall be measured across the 150 ohm resistor. Results shall comply with 3.12.5.

4.6.6 Microphone endurance test.- A warble frequency signal of 800 to 1250 c.p.s. at a voltage of 25 millivolts (r.m.s.) shall be applied across the microphone voice coil for an uninterrupted period of 8 hours with the microphone in an ambient room temperature of 72 ±5° F. Results shall comply with 3.12.6.

4.6.7 Switch endurance test.- The microphone control circuits shall be connected to a 115 volt 60 c.p.s. supply through suitable resistance and inductance to cause a current of 0.5 amperes at 50 percent power factor to flow in the control circuit when the press-to-talk switch is closed. The press-to-talk switch shall be operated 100,000 times to determine compliance with 3.12.7.

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4.6.8 Cable assembly strength test.- A microphone holder assembly of the type specified in 3.11 shall be mounted on a firm surface. The microphone under test shall be complete with cable and plug and shall be locked into the microphone holder assembly. A force of 20 pounds shall be applied to the connector plug in the direction away from the microphone case and along a line which is coincident with the axis of the portion of cable that is clamped in the microphone case. Results shall comply with 3.12.8.

4.6.9 Dielectric strength test.- Dielectric strength between electrical circuits and between electrical circuits and external metal parts of the microphone shall be tested in accordance with MIL-I-983. Results shall comply with 3.12.9.

4.6.10 Insulation resistance.- The insulation resistance of the electrical circuits of the microphone shall be measured at a room temperature of 72 ±5° F. and at a relative humidity of approximately 50 percent using a voltage of 50 or more volts d.c. to determine compliance with 3.12.10.

4.6.11 Extended storage test.- The microphone shall be subjected to the extended storage test specified in MIL-I-983. Results shall comply with 3.12.11.

4.6.12 Salt spray test.- The microphone shall be subjected to a salt spray test performed in accordance with method 101 of MIL-STD-202. The salt solution concentration shall be 20 percent. Length of the test shall be 96 hours. Results shall comply with 3.12.12.

4.6.13 Gunblast test.- The microphone shall be subjected to the simulated gunblast test specified in MIL-I-983. Results shall comply with 3.12.13.

4.6.14 Random drop test.- The microphone shall be subjected to 10 random drops from a 6 foot height onto a concrete surface. Results shall comply with 3.12.14.

4.6.15 Vibration test.- The microphone shall be placed in a microphone holder assembly of the type specified in 3.11 which is mounted on a simulated bulkhead. The microphone shall then be subjected to the vibration test for vital equipment specified in MIL-I-983 except that the microphone circuits shall not be energized during the test. Results shall comply with 3.12.15.

4.6.16 Shock test.- The microphone shall be placed in a microphone holder of the type specified in 3.11 which is mounted on a simulated bulkhead. The microphone shall then be subjected to the type A, HI shock test specified in MIL-S-901 except that the microphone circuits shall not be energized during the test. Results shall comply with 3.12.16. Microphones which have been subjected to the shock test shall not be installed aboard ship.

4.6.17 Frequency response after environmental tests.- At the conclusion of the series of environmental tests specified in 4.6.11 through 4.6.16, the frequency response test specified in 4.6.1.4 shall be repeated. Results shall comply with 3.12.17.

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation.-

5.1.1 Microphones.-

5.1.1.1 Preservation and packaging.- Preservation and packaging shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation and may conform to the supplier's commercial practice when such meets these requirements.

5.1.1.2 Packing.- Packing shall be accomplished in a manner which will insure acceptance by common carrier and will afford protection against physical and mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation and may conform to the supplier's commercial practice when such meets these requirements.

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5.1.1.3 Marking.- Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the supplier's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, manufacturer's name and destination.

5.2 Domestic shipment and storage or overseas shipment.- The requirements, and levels of preservation, packaging, packing and marking for shipment shall be specified by the procuring activity (see 6.2).

(5.2.1 The following provides various levels of protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made:

5.2.1.1 Preservation and packaging, packing and marking.- The equipment and accessories, and technical publications shall be preserved and packaged levels A or C; packed level A or B as specified and marked in accordance with MIL-E-17555.)

6. NOTES

6.1 Intended use.- The microphones covered by this specification are intended for use with intercommunication and announcing equipment aboard Naval ships.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Quantity of items required.
(c) Type required (see 1.2).
(d) Whether a microphone holder assembly is required with each microphone (see 3.2).
(e) Preservation and packaging, packing and marking, if other than as specified in 5.1 (see 5.2).
(f) Government laboratory to which preproduction samples shall be submitted (see 6.3.1).

6.3 Preproduction.- Invitations for bid should provide that the Government reserves the right to waive the requirement for preproduction samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products who wish to rely on such production or test must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

6.3.1 The two preproduction samples (see 3.1) shall be submitted to a Government laboratory specified in the contract or order (see 6.2).

6.4 CHANGES FROM PREVIOUS ISSUE.- THE EXTENT OF CHANGES (DELETIONS, ADDITIONS, ETC.) PRECLUDE THE ANNOTATION OF THE INDIVIDUAL CHANGES FROM THE PREVIOUS ISSUE OF THIS DOCUMENT.

Preparing activity:
Navy - SH
(Project 5830-N065SH)

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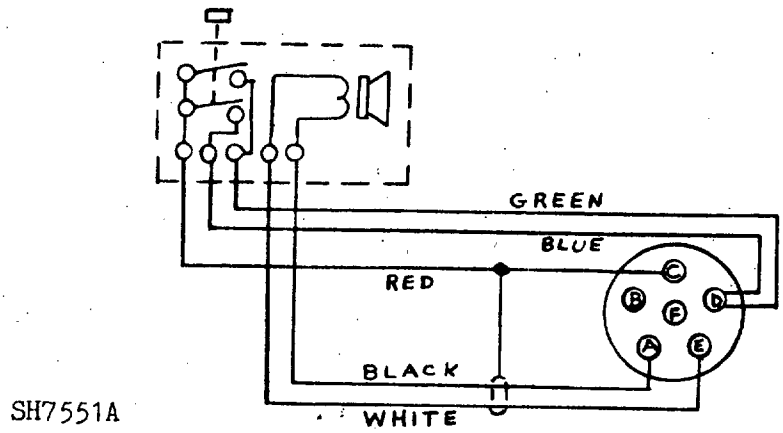


Figure 1 - Electrical circuit for type M132/SIC microphone.

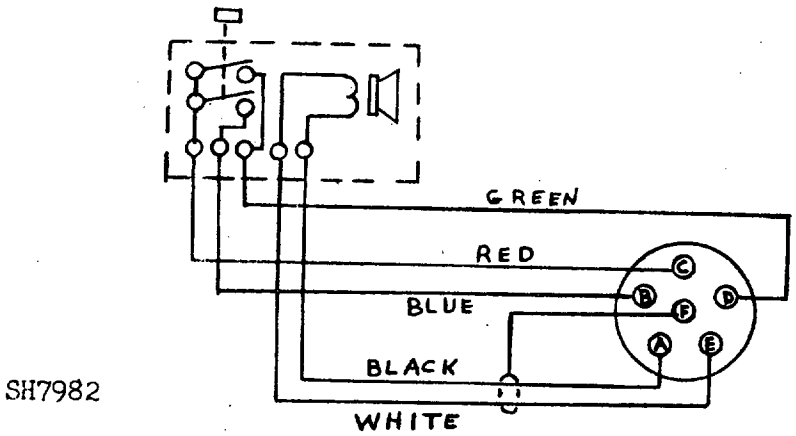


Figure 2 - Electrical circuit for type M124/SIC and M125/SIC microphones.

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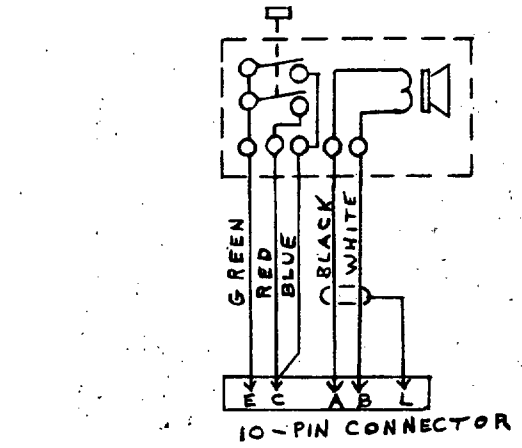


Figure 3 - Electrical circuit for type M131/WIC mircophone.

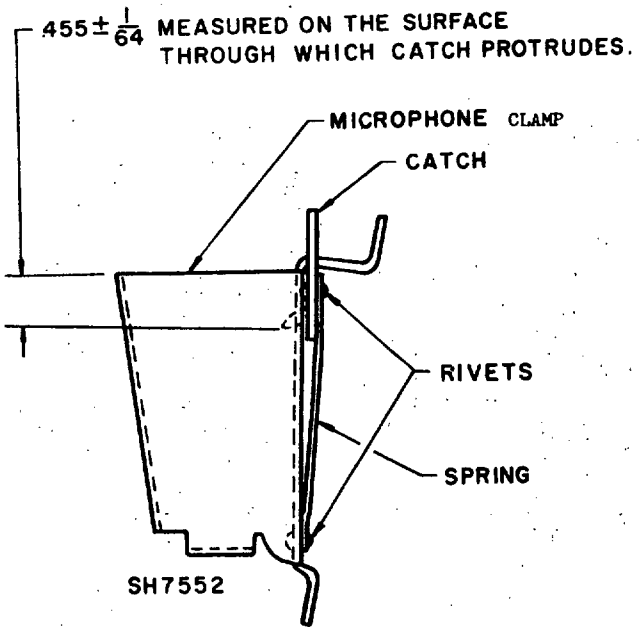


Figure 4 - Microphone holder assembly.

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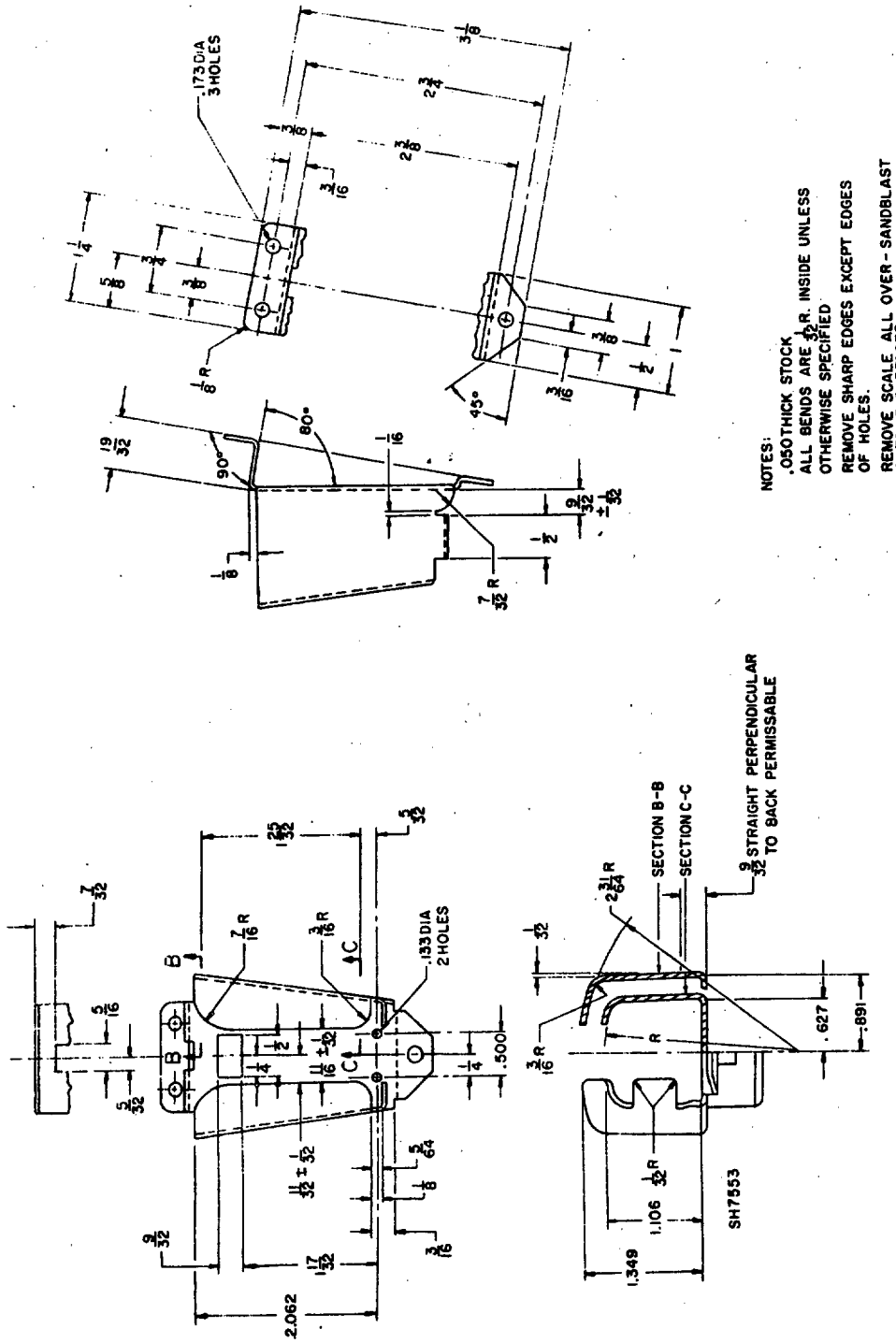


Figure 5 - Microphone clamp.

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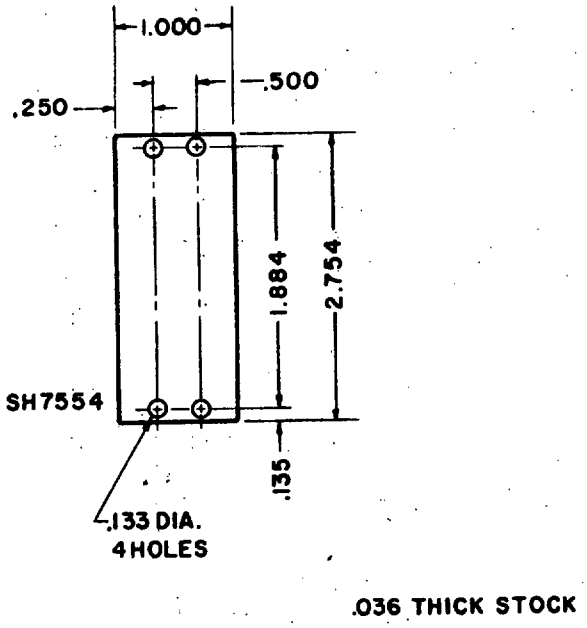


Figure 6 - Spring.

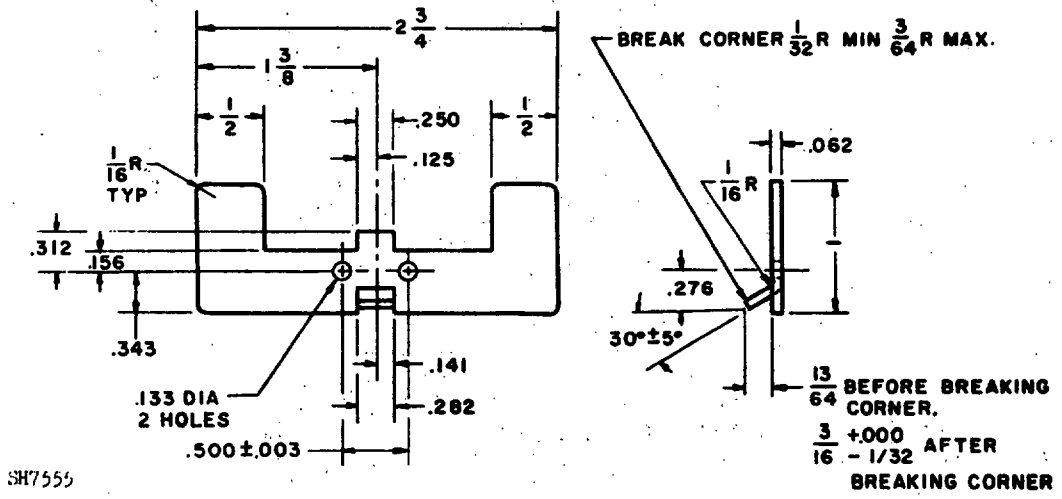


Figure 7 - Catch.

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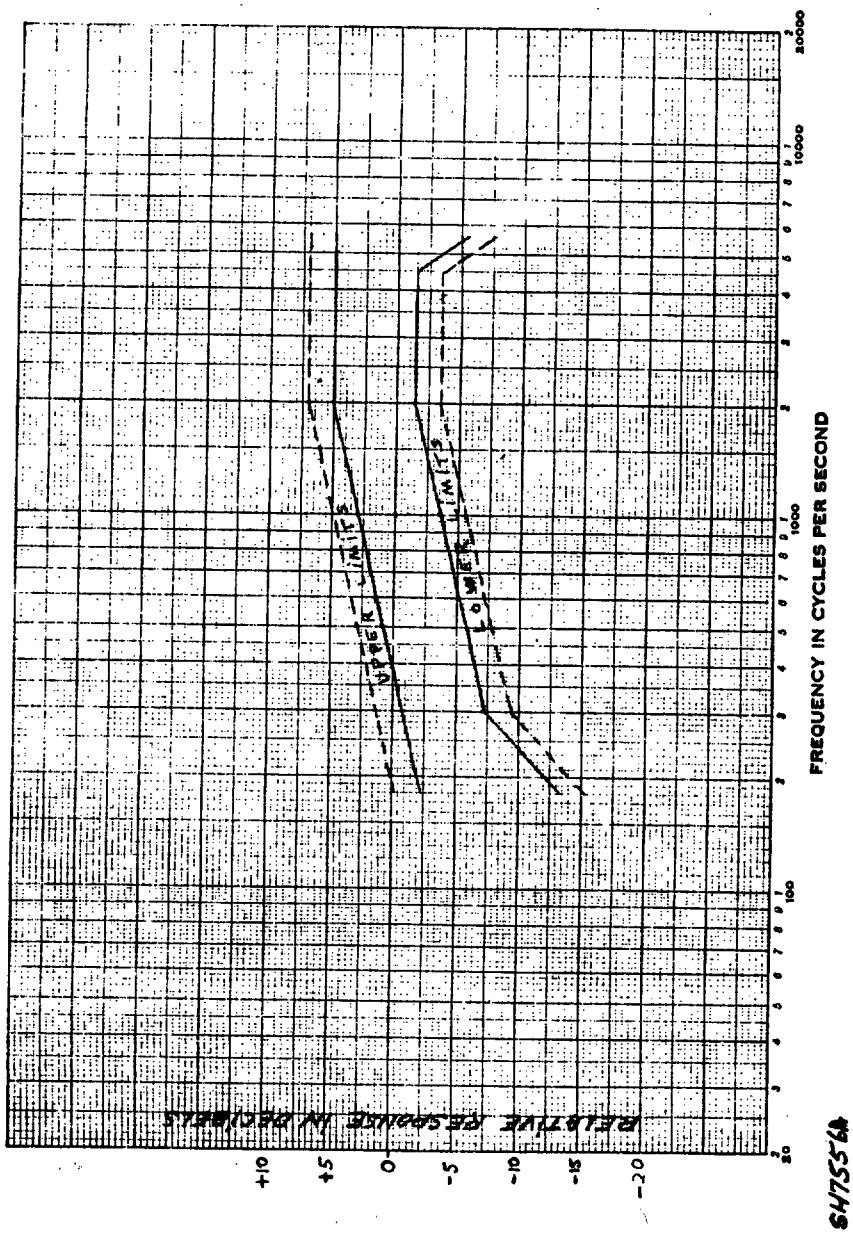
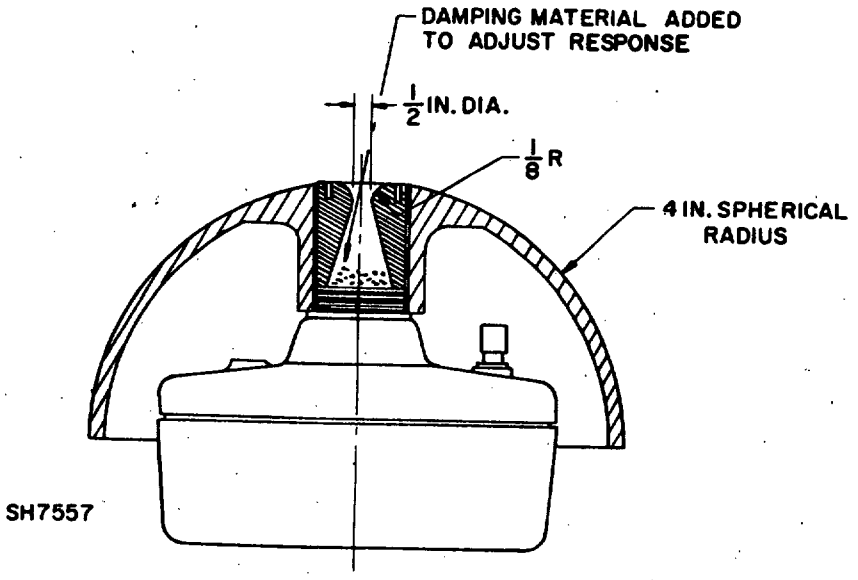


Figure 8 - Frequency response limits for microphones.

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CONSTRUCTION OF THE HEMISPHERE SHALL BE SUCH AS TO AVOID DISTURBANCE OF THE SOUND FIELD DUE TO ITS VIBRATION, OR SOUND ABSORPTION.

Figure 9 - Artificial voice.

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OMB Approval
No. 22-R255

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A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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